

Key Features

- The IC control circuit and the LED point light source share a single power supply.
- Operating current per channel is 12 mA.
- The control circuit and RGB chip are integrated into a single 2020-package component, forming a complete externally controlled pixel.
- A built-in signal shaping circuit ensures that signals received by any pixel are shaped before output, preventing the accumulation of signal distortion along the line.
- Built-in power-on reset and power-off reset circuits.
- Each pixel's three primary colors can achieve 256 levels of brightness, enabling true-to-life color display with 16,777,216 colors.
- Port scanning frequency: 2 kHz/s.
- Serial cascading interface enables data reception and decoding via a single signal line.
- At a refresh rate of 30 frames per second, the number of cascaded pixels is no less than 1,024.
- Data transmission speed can reach 800 Kbps.
- High color consistency and excellent cost-performance ratio.
- **Resistant to reverse polarity connection without damage.**
- **Static power consumption is less than 1 μ A, and it supports a 3.3 V power supply.**

Main Application Areas

- Consumer electronics.
- LED lighting and illumination.
- Computers and peripherals, gaming equipment, and various electrical appliances.

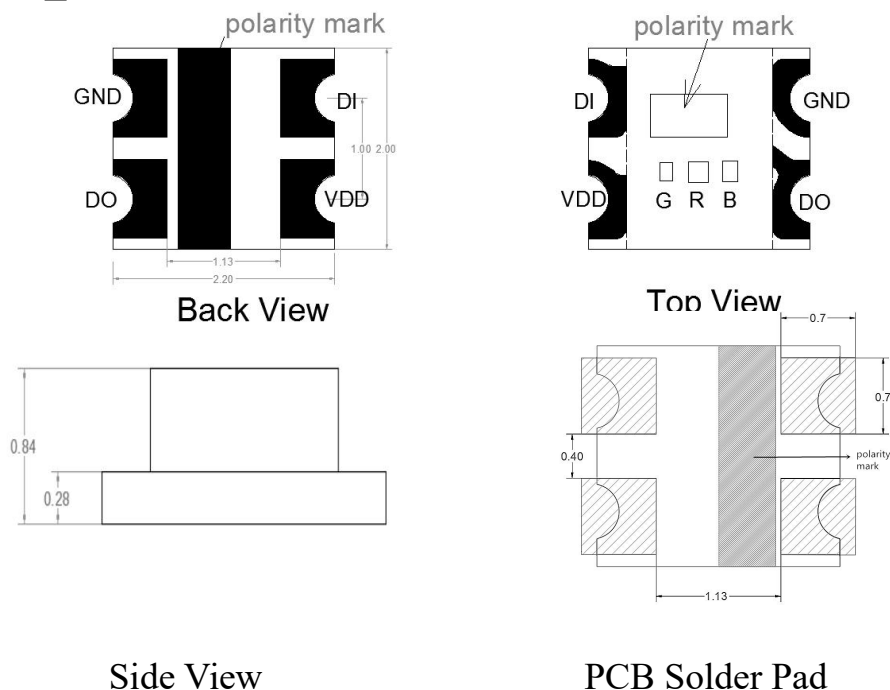
Product Overview

The WS2812B-2020-V6 is a smart externally controlled LED light source that integrates both the control circuit and the light-emitting circuit into a single unit; Its design utilizes the latest molding packaging technology, integrating the IC and light-emitting chip into a 2020-sized package, with each component functioning as a single pixel; the pixel contains a smart digital interface, data latch, signal shaping, amplification, and drive circuitry, as well as a high-precision internal oscillator and a programmable constant-current control section, effectively ensuring high color consistency of the pixel's light.

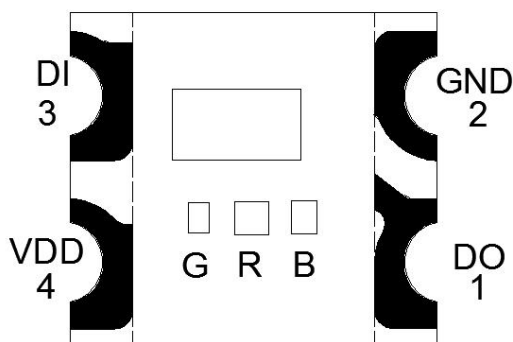
The data protocol employs a single-wire zero-return code communication method. After power-on reset, the DIN port receives data transmitted from the controller. The first 24-bit data received is extracted by the first pixel and sent to the internal data latch. The remaining data is shaped and amplified by the internal shaping circuit before being forwarded via the DO port to the next cascaded pixel; with each pixel, 24 bits of the signal are reduced. The pixel employs automatic shaping and forwarding technology, ensuring that the number of pixels that can be cascaded is not limited by signal transmission but only by signal transmission speed requirements; With a port scanning frequency of up to **2 kHz**, no flickering occurs even when captured by a high-definition camera, making it ideal for use in high-speed moving products; With a **RESET** time exceeding 280 μ s, interrupts will not cause erroneous resets, allowing support for lower-frequency, lower-cost MCUs; The LEDs offer advantages such as low-voltage drive, environmental friendliness and energy efficiency, high brightness, wide beam angle, excellent uniformity, low power consumption, and ultra-long lifespan. By integrating the control circuit onto the LED, the circuit becomes simpler, more compact, and easier to install.

Mechanical and Recommended Pad Dimensions (in mm)

Tolerance: ± 0.15 mm



Pin Arrangement



Pin Function

No.	Symbol	Pin Name	Function Description
1	DO	Data Output	Control data signal output
2	GND	Ground	Signal ground and power ground
3	DI	Data Input	Control Data Signal Input
4	VDD	Power Supply	Power Supply Pin

Maximum Rated Values ($T_A = 25^\circ\text{C}$, $V_{SS} = 0\text{V}$)

Parameters	Symbol	Range	Unit
Supply Voltage	VDD	+3.3 to +5.3	V
Logic Input Voltage	VI	-0.3 V to VDD +0.7	V
Operating Temperature	Topt	-40 to +85	°C
Storage Temperature (Bulk Material)	Tstg	-40 to +105	°C
Storage Temperature (Coils)	Tstg	-40 to +70	°C

Electrical Parameters ($T_A = 25^\circ\text{C}$, $V_{(DD)} = 5\text{V}$, $V_{SS} = 0\text{V}$)

Parameter	Symbol	Minimum	Typical	Maximum	Unit	Test Conditions
Quiescent Current	Io	—	—	1	uA	No signal transmission; RGB channels are off
Input current	II	—	—	±1	μA	$V_I = V_{DD}/V_{SS}$
High-level input	V _{IH}	0.55 VDD	—	VDD + 0.7 V	V	D _{IN} , SET
Low-level input	V _{IL}	-0.3V	—	0.7V	V	D _{IN} , SET

Switching Characteristics ($T_A = 25^\circ\text{C}$, $V_{(DD)} = 5\text{V}$, $V_{SS} = 0\text{V}$)

Parameters	Symbol	Minimum	Typical	Maximum	Unit	Test Conditions
Oscillation Frequency	Fosc	—	800	—	kHz	—
Transmission delay time	t _{PLZ}	—	—	300	ns	CL=15 pF, DIN→DO _{UT} , RL=10 kΩ
Fall time	t _{THZ}	—	—	120	μs	CL=300 pF, OUT _R /OUT _G /OUT _B

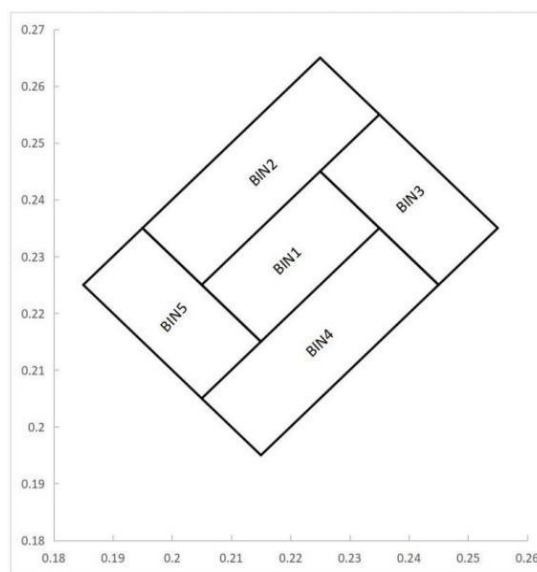
Data Transfer Rate	F_{MAX}	600	—	—	Kbps	50% duty cycle
Input capacitance	C_I	—	—	15	pF	—

LED Specifications

Parameter	Symbol	Color	Minimum Value	Typical Value	Maximum	Unit	Operating Current
Luminous Intensity	IV	Red	170	340	680	mcd	12 mA
		Green	425	850	1700		
		Blue	130	260	520		
Wavelength	λ_d	Red	615	—	625	nm	12 mA
		Green	515	—	525		
		Blue	465	—	475		
Color Mix	XY	White	—	x: 0.21 y: 0.23	—	/	36 mA
	CCT	White	25,000	—	—	K	

BIN Classification

BIN1	X	0.215	0.205	0.225	0.235
	Y	0.215	0.225	0.245	0.235
BIN2	X	0.205	0.195	0.225	0.235
	Y	0.225	0.235	0.265	0.255
BIN3	X	0.245	0.225	0.235	0.255
	Y	0.225	0.245	0.255	0.235
BIN4	X	0.215	0.205	0.235	0.245
	Y	0.195	0.205	0.235	0.225
BIN5	X	0.205	0.185	0.195	0.215
	Y	0.205	0.225	0.235	0.215

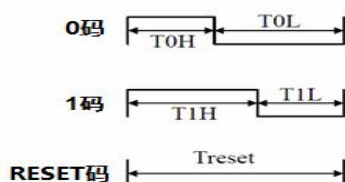


Data Transfer Time

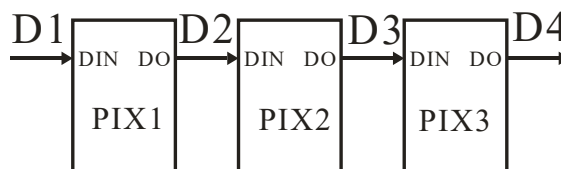
T0H	Code 0, High-level duration	220 ns–380 ns
T1H	Bit 1, High-level duration	580 ns–1 μs
T0L	Code 0, Low-level duration	580 ns–1 μs
T1L	Code 1, Low-level duration	580 ns to 1 μs
RES	Per frame, low-level duration	280 μs or more
Data cycle: $T0H + T0L, T1H + T1L \geq 1.25 \mu s$		

Timing Waveform Diagram

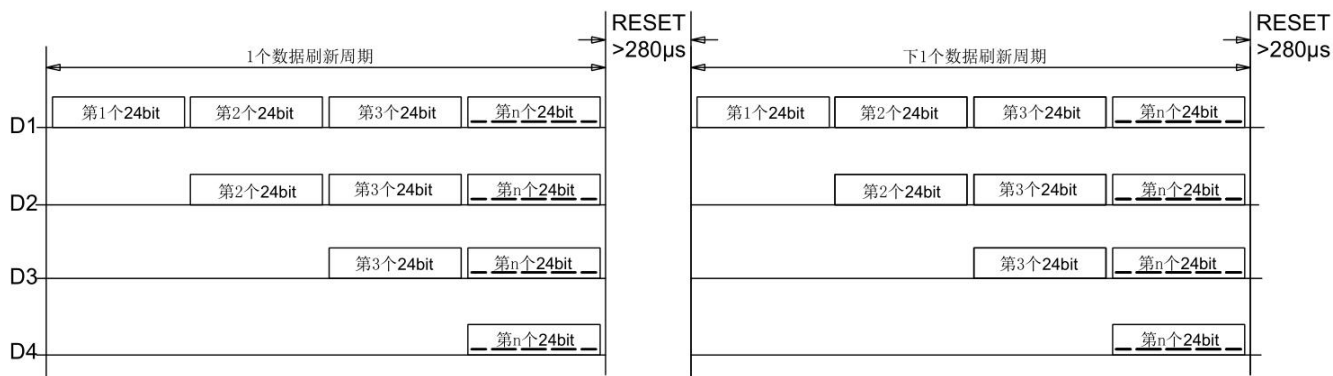
Input Pattern:



Connection Method:



Data Transmission Method



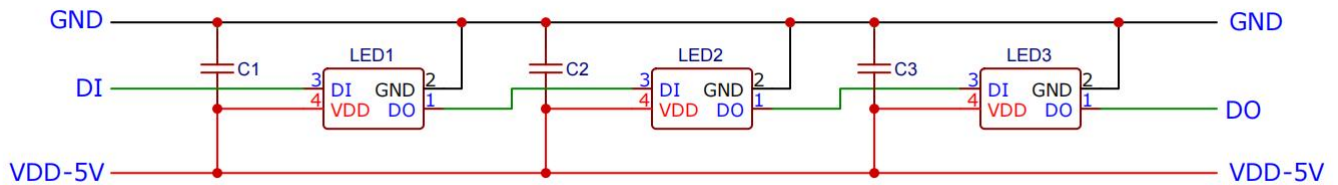
Note: D1 represents data transmitted from the MCU, while D2, D3, and D4 represent data automatically shaped and forwarded by the cascaded circuit.

24-bit Data Structure

G7	G6	G5	G4	G3	G2	G1	G0	R7	R6	R5	R4	R3	R2	R1	R0	B7	B6	B5	B4	B3	B2	B1	B0
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Note: Send the most significant bit first, and transmit data in the GRB order.

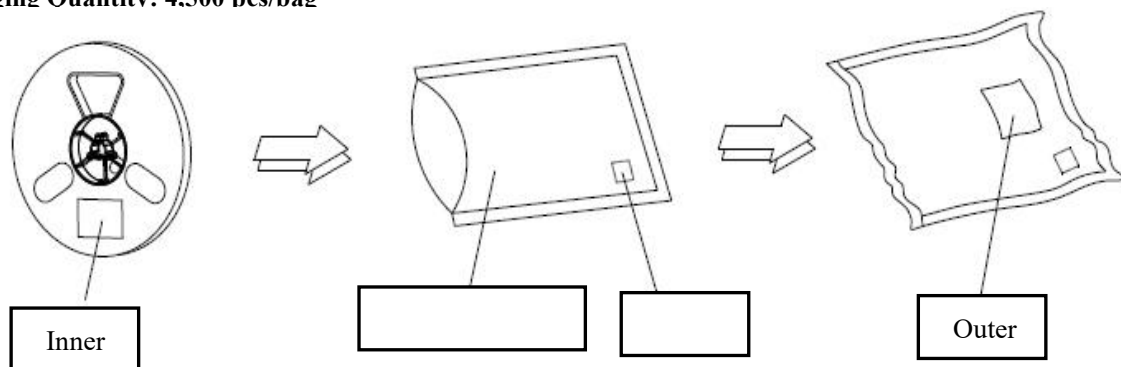
Typical Application Circuit



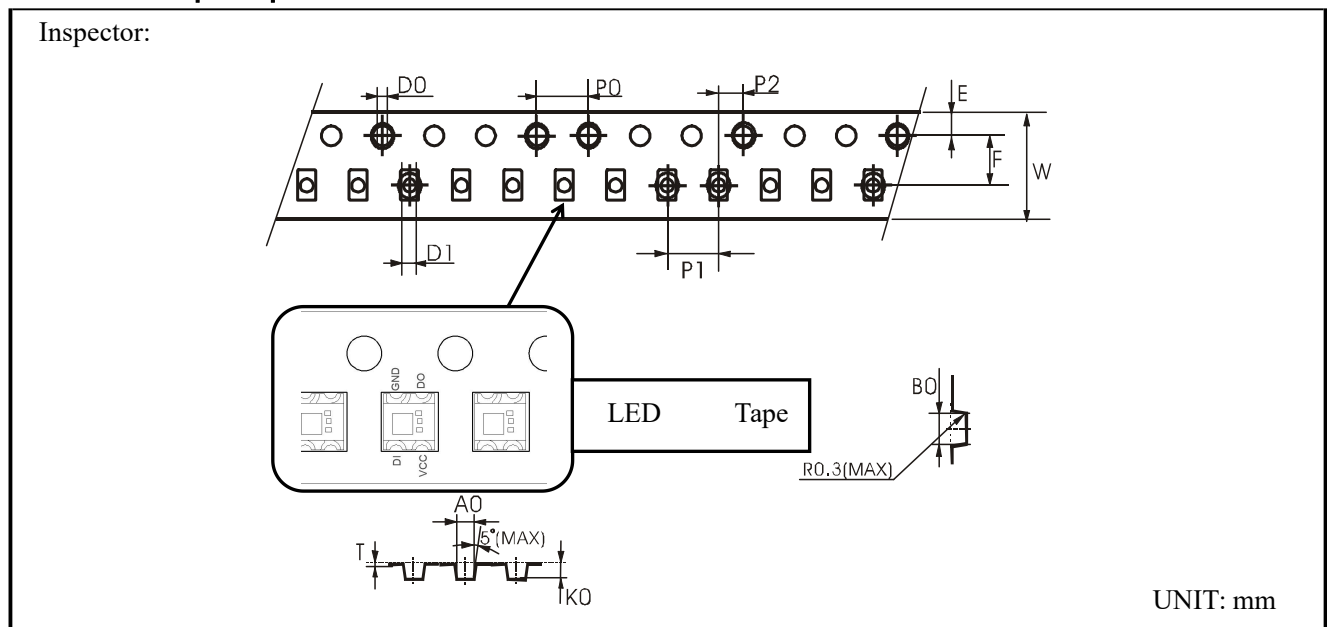
Where C_n is the filter capacitor for the LED's VDD pin, typically set to 100 nF.

Packaging Specifications

Packaging Quantity: 4,500 pcs/bag



Carrier Tape Specifications

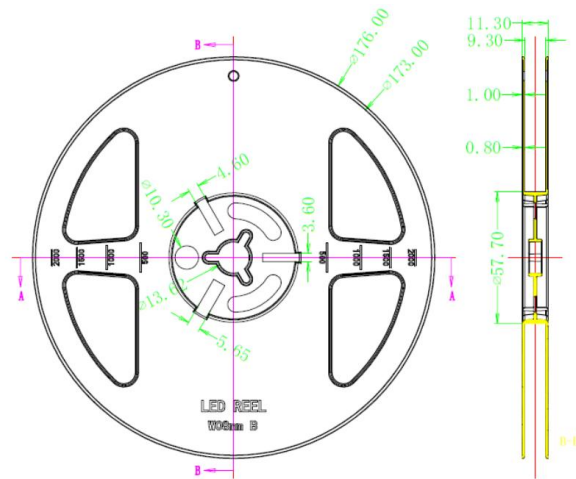


Carrier Tape Test Reports

	A0	B0	K0	P0	P1	P2	T	E	F	D0	D1	W
SPEC	2.20	2.40	1.01	4.00	4.00	2.00	0.18	1.75	3.50	1.50	1.00	8.00

Reel Dimensions

Unit: mm



Precautions for Surface-Mount LEDs

1. Description:

Generally, LEDs are used in the same way as other electronic components. To help customers make the best use of Huacaiwei's LED products, please refer to the following LED protection precautions.

2. Precautions:

2.1. Dust and Cleaning

The surface of the LED is encapsulated with modified epoxy resin, which provides excellent protection for the LED's optical system and aging resistance. Epoxy resin tends to attract dust, so maintain a clean work environment. While a limited amount of dust on the LED surface will not affect its brightness, you should still avoid allowing dust to settle on the LED surface. Use LEDs immediately after opening the packaging. Assemblies that have been installed with LEDs should be stored in clean containers. When cleaning the LED surface, solutions such as triamine or acetone may cause the surface to dissolve; therefore, do not use any solvents with dissolving properties to clean the LEDs. Isopropyl alcohol may be used, but you must verify that the cleaning solution will not dissolve the LED before use. Please do not clean LEDs using ultrasonic methods. If ultrasonic cleaning is necessary for a product, evaluate parameters that may affect the LEDs, such as ultrasonic power, drying time, and assembly conditions. A test run must be conducted before cleaning to confirm whether it will affect the LEDs.

2.2. Moisture-Proof Packaging

LEDs are moisture-sensitive components. They are packaged in aluminum foil pouches to prevent them from absorbing moisture during transportation and storage; desiccant is included in the pouch to absorb moisture. If an LED absorbs moisture, the moisture will evaporate and expand as the LED passes through the reflow soldering process, which may cause the adhesive to separate from the substrate and damage the LED's optical system. For this reason, moisture-proof packaging is designed to prevent moisture from entering the bag; however, the protection typically lasts only 1 to 2 months. The moisture sensitivity level (MSL) for this product is **5a**. During SMT, please refer to the material moisture sensitivity level (MSL) definitions specified in IPC/JEDEC J-STD-020 for MSL control.

Moisture Sensitivity Level	Shelf Life After Opening	
	Time	Conditions
LEVEL 1	Unlimited	$\leq 30^{\circ}\text{C}/85\%\text{RH}$
LEVEL 2	1 year	$\leq 30^{\circ}\text{C}/60\%\text{RH}$

LEVEL 2a	4 weeks	$\leq 30^{\circ}\text{C}/60\%\text{RH}$
LEVEL 3	168 hours	$\leq 30^{\circ}\text{C}/60\%\text{RH}$
LEVEL 4	72 hours	$\leq 30^{\circ}\text{C}, 160\%\text{RH}$
LEVEL 5	48 hours	$\leq 30^{\circ}\text{C}/60\%\text{RH}$
LEVEL 5a	24 hours	$\leq 30^{\circ}\text{C}/60\%\text{RH}$
LEVEL 6	Ready to use upon removal	$\leq 30^{\circ}\text{C}/60\%\text{RH}$

2.3 SMT Placement Instructions:

2.3.1 Remove the LEDs from their packaging prior to SMT and place the entire roll in an oven for dehumidification and drying (bake at $70 - 75^{\circ}\text{C}$ for ≥ 24 hours);

2.3.2 The time from when the product is removed from the oven until high-temperature soldering is completed (including multiple reflow soldering cycles, tin dipping, wave soldering, and heat-related repairs) must be kept within 24 hours (under conditions of $T < 30^{\circ}\text{C}$ and $\text{RH} < 60\%$);

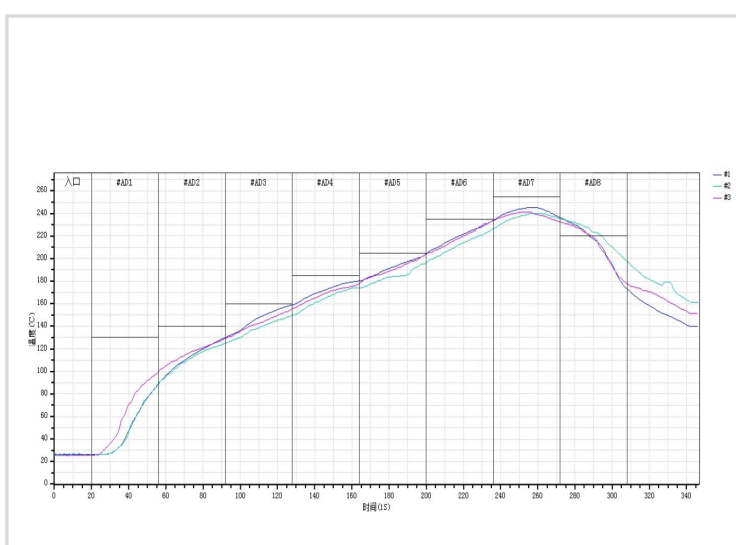
2.3.3 SMT assembly of LEDs onto PCBs after solder paste printing should be completed as soon as possible; it is recommended that this be done within 1 hour;

2.3.4 Loose LEDs — such as production surplus, machine scrap, or repair parts — that have been exposed to air for an extended period should not be used directly; it is recommended to dehumidify and dry them before use. Baking for entire rolls: $70 - 75^{\circ}\text{C}^*$ for ≥ 24 hours; or baking for loose LEDs: 120°C for 4 hours.

3. Soldering

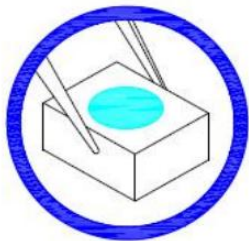
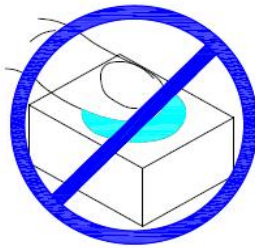
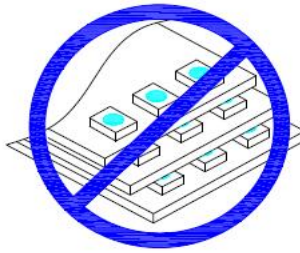
LEDs used in surface-mount applications must comply with the JEDEC J-STD-020C standard. As a general guideline, it is recommended to follow the soldering temperature profile recommended by the solder paste manufacturer, or to use the soldering temperature profile recommended by our company as follows.

Temperature Profile	Range
$30^{\circ}\text{C} - 150^{\circ}\text{C}$ Preheat	$1 - 4^{\circ}\text{C/s}$
$30^{\circ}\text{C} - 150^{\circ}\text{C}$ Preheat	60 - 120 s
$150^{\circ}\text{C} - 200^{\circ}\text{C}$	$0 - 3^{\circ}\text{C/s}$
$150^{\circ}\text{C} - 200^{\circ}\text{C}$	60 - 120 s
Liquid-phase	217°C
Peak temperature	245°C
Reflow slope	$0 - 3^{\circ}\text{C/s}$
Reflow Soldering Time	45 - 90 s
Cooling Rate	-4 to 0°C/s
Dwell time from room	< 6 min



Note: 1. All temperatures listed above refer to those measured on the upper surface of the package body

4. Precautions During Product Assembly

1. Use appropriate tools to grasp the package from the side	2. Do not press the surface of the gel directly with your hands or sharp metal objects, as this may damage the internal circuitry	3. Do not stack the module materials on top of each other, as this may damage the internal circuitry	4. Do not use in acidic environments where the pH is less than 7
			

Document Change Log

Version Number	Status	Summary of Changes	Revision Date	Reviser	Approver
V1.0	N	New	October 21, 2025	Chen Yongzhao	Yin Huaping
V1.1	A	Add appropriate filtering capacitors to the application circuit	December 1, 2025	Ouyang Yu	Yin Huaping
V1.2	M	Optimized application circuit diagram	March 18, 2026	Chen Yongzhao	Yin Huaping
V1.3	M	Changed the logo; optimized the application circuit diagram	July 10, 2026	Chen Yongzhao	Yin Huaping
V1.4	M	LED Characteristics; Added “Signal Transmission Rate” description and dimensional tolerances	July 28, 2026	Chen Yongzhao	Yin Huaping